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SECTION 1

Cognition and Metacognition

- **Five failure modes:** Production agents often emit confident wrong answers, repeat failed searches, cling to obsolete plans, hallucinate past missing data, or flatten compositional questions into one lookup; each maps to a structural module rather than a prompt fix.
- **Primitives versus architecture:** Chain-of-thought, ReAct, trees, and reflection are skills; architecture chooses which skill to deploy, watches whether it works, and reroutes when evaluation signals stagnation or contradiction.
- **Cognition defined:** Agent cognition is the fidelity of the internal task model—decomposition, dependency ordering, novel tool chaining, and mid-run plan revision when new facts contradict earlier assumptions.
- **Metacognition defined:** Metacognition covers confidence that tracks correctness, stagnation detection before iteration limits burn out, and behavior that changes when retrieval and memory no longer cover the question.
- **Theoretical foundations:** Minsky favors a society of narrow modules; Baars supplies a global workspace where signals broadcast; Kahneman motivates a fast path for trivial queries and a full cognitive cycle for ambiguous or contradictory work.

SECTION 2

Cognitive Architecture

- **Cognitive workspace:** A typed shared state holds task representation, strategy, findings, confidence, execution history, and attention signals such as contradiction, stagnation, or knowledge gaps.
- **Perception and planning:** Perception classifies task type and complexity before any tool call; planning picks direct, decompose, explore, or hypothesis-test strategies and can lean on memory hits from prior cases.
- **Execution and evaluation:** Execution wraps tool output in scored findings; evaluation is real-time metacognition that judges evidence quality, consistency, progress, and whether to continue, replan, escalate, or stop.
- **Attention routing:** Deterministic routing implements system 1 fast respond, default perceive-plan-execute-evaluate cycles, meta-plan after stagnation, replan on contradiction, and escalate when confidence collapses.
- **Memory via MCP:** The official MCP memory server stores entities, relations, and observations; proactive retrieval before planning and experience recording after evaluation turn sessions into cumulative institutional knowledge.

SECTION 3

Building and Running the Cognitive Agent

- **Cognitive loop:** The outer agentic loop still governs iteration and exit; inside each iteration, attention may run multiple cognitive steps until signals clear or the task completes.
- **Confidence gate:** Presentation to the user is gated on workspace confidence, unresolved contradictions, and declining confidence trends, favoring more gathering or explicit uncertainty over polished guesses.
- **Stagnation pivot:** Overlapping findings or flat confidence trends raise a stagnation signal, record the failed strategy, and route to meta-planning so the planner cannot replay the same approach.
- **Knowledge boundary:** Combined retrieval quality, memory coverage, and confidence classify whether the agent is within, at the edge of, or outside known territory, triggering low-confidence handling when coverage is thin.
- **Emergent behaviors:** Module interaction produces curiosity after knowledge gaps, adaptive persistence after failure, selective depth on simple familiar queries, and graceful degradation when budgets or confidence run out.

SECTION 4

Measuring Cognitive Capability

- **Diagnostic tests:** Five practical probes mirror the production failure modes—metadata mistaken for content, repeated queries, plans that ignore new facts, hallucination without coverage, and flat single-step answers to compositional questions.
- **Efficiency and calibration:** Track cognitive efficiency (steps versus optimal), metacognitive calibration (confidence versus correctness), adaptation rate after stagnation, and knowledge-boundary accuracy for false certainty versus false caution.
- **Before-and-after studies:** Compare a flat ReAct agent with the cognitive stack on the same queries, then contrast an empty knowledge graph against a graph populated after many tasks to see memory compound.
- **Generalization path:** Swapping perception labels, execution MCP servers, and planning strategies reuses the same workspace, attention, evaluation, and gating patterns across domains without rescaling the base model alone.